

Tuğrul Ağrikli

Software Engineer · Machine learning and computer vision · Planning and simulation · Linux systems

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PROFILE

Software engineer (BSc Computer Science and Engineering, June 2026) who builds whole systems, from embedded C and Linux desktop software to GPU inference, and measures them. In machine learning and computer vision: real-time depth inference on TensorRT at a steady 60 fps, and 202 logged experiments on learning rules that work without backpropagation. At Parkolay, built the deterministic planner that the company switched on at a live five-lift car park in September 2026, after a reinforcement-learning approach missed its target. Looking for machine learning, computer vision, planning or Linux systems roles.

EXPERIENCE

Software Engineer, Planning and Simulation (R&D), Parkolay, İstanbul · Apr 2026 to Aug 2026

Automated parking system manufacturer (AS/RS), 50 to 200 staff

- Designed and built OOSSolver, the deterministic planner that schedules every lift and shuttle move in a multi-lift automated car park. After handover, the company connected it to its control software and switched it on at a live five-lift car park in September 2026.
- Delivered it with an adapter (120 tests) that turns the company's facility model and live PLC data into planner input. It reports problems instead of crashing, tells missing data apart from equipment known to be down, and found six out-of-range rows in a reference configuration that the company's own validator had missed.
- Tested it first in a discrete-event simulator built for the job, against seven test gates of its own. In a 30-day run on a five-lift test layout, with evening demand at 1.7 times what the layout can drain, it handled 10,814 arrivals and 10,858 departures without ever getting stuck. The long runs exposed a starvation loop and a three-way deadlock, both traced on exact replays and fixed under regression tests.
- Began with a single-lift A* planner and its C++20 port (24.7 times faster than Python, identical results on 106 tasks). For the multi-lift plant, tried reinforcement learning (PPO with a graph-attention policy) and dropped it for the deterministic design when it stalled below its target.

Write-up with a 75-second demo: zerobitfulladder.github.io/#work/oossolver

Independent research and engineering, self-directed, İstanbul · Feb 2026 to present

- Built Zenith, a log of 202 short experiments, mostly without backpropagation, on three questions: can a system learn from one example at a time using only local updates, keep what it knew when new things arrive, and both recognise and generate from the same memory? Covered 13 directions, including local learning rules, sparse codes, continual learning, sequence memory, generation and motor control; each experiment keeps its code, baselines, numbers and a written reading, failures included, with the expected result written down before key runs.
- Built AxonForge, a PySide6 node-graph framework for prototyping algorithms that change step by step: graphs may contain loops, nodes are plain Python classes that reload while running, and parameters and internal state can be watched and tuned live. Zenith's early experiments ran on it.
- Built Breadcrumbs (write-up), a local tool for studying research literature (FastAPI, React, SQLite). Its LLM assistant has 27 tools across five model providers, reads the library through a separate read-only database connection, and can propose papers but never add them.

R&D Intern, ParkMatrix team, Parkolay, İstanbul · Jul 2025 to Sep 2025

- Built the AprilTag camera localisation kept in the final prototype of ParkMatrix, the company's self-driving pallet robot (AMR), along with a USB serial protocol (CRC checks, ACK/NACK, automatic reconnect), CANopen servo control, and a Modbus RTU logger for laser-scanner signals.
- Wrote Raspberry Pi Pico encoder firmware (MicroPython, then C on the PIO). The sampling limits it exposed contributed to the team's switch to custom motor-driver boards.
- After the internship, audited a vendor's car-positioning model and showed its accuracy could not be trusted: near-duplicate frames of 312 scenes sat on both sides of the split, and a wiring bug meant the model trained only on the validation set.

Freelance Software Engineer, *Upwork and direct clients, remote* · 2020 to 2023

- Completed 12 jobs on Upwork. The longest, a Raspberry Pi information display for a client in Sweden, combined weather-station, pool-sensor and energy-meter data with a live camera feed and ran unattended around the clock (five months, five-star review).
- Built a service that records calls automatically on Librem 5 Linux phones (code; ModemManager over D-Bus, PulseAudio, GStreamer), and an email pipeline for a business client that paired each password email with its protected archive, about 20,000 messages a week.
- Built research software for two university groups: for an Istanbul Technical University architecture study, a Rhino/Grasshopper to MATLAB fuzzy-logic bridge, [acknowledged in Architectural Science Review](#) (2023); for Sabancı University's magnetar group, a data platform and public search site (MySQL, React), still online.

EDUCATION

BSc Computer Science and Engineering (English-medium), *Sabancı University, İstanbul* · June 2026

- Graduation project (write-up): tested whether a camera and a MobileNetV3 classifier (transfer learning) could replace laser scanners for spotting a car that overhangs its pallet, on 20,000 deduplicated frames from a live car park. On a random 20% split it reached 98.9% accuracy and 98.1% recall on occupied edges, the safety-critical class; a second camera or a later week is still untested. Model-assisted labelling cut about 200 hours of work to two; Blender synthetic pre-training traded fewer false alarms for more missed cars.
- Machine Learning course: the team's classifier (text embeddings with XGBoost) ranked first in the final round of the class competition (code).
- Relevant courses: Introduction to Data Science, Linear Algebra, Artificial Intelligence, Statistical Modelling, Database Systems, Computer Networks, Cryptography.

SELECTED PROJECTS · write-ups and videos at zerobitfulladder.github.io

VitureBD. XR, 2026. Shows the whole Linux desktop in live 3D on XR glasses at a steady 60 fps on a 6 GB laptop GPU. The depth model runs as a TensorRT FP16 engine (about 120 fps, against 67 in PyTorch), zero-copy CUDA-OpenGL interop keeps frames on the GPU, and profiling the 16.7 ms frame budget per stage led to cutting the stereo warp from 4.63 to 0.86 ms (write-up).

Monitor-anchored AR. XR, 2026. World-locked AR on VITURE XR glasses under Linux, anchored to a desktop monitor. Marker tracking, hand-eye calibration and per-eye display calibration kept virtual objects within about 1 to 2 cm, judged by eye. Worked around three undocumented faults in the closed SDK (a pose call that overran its buffer, crashes with two instances, and USB drops after power saving), and a Blender add-on streams scenes into the glasses live.

indigoshell. Open source, 2026. A desktop shell for X11 in daily use: bar, panels, launcher, notification daemon and system tray in one Python process (XCB, Skia on EGL, D-Bus). Repainting only the regions that changed cut idle CPU from 25.8% to 4.5% of a core, with output matching full repaints byte for byte over 360 frames; 13 GitHub stars (write-up).

Eileen. LLM application, 2026. A virtual live-streamer whose feelings are computed, not prompted. One language model reads each chat message, deterministic rules turn it into emotions that rise and fade, and a second model, told only how she acts and never what she feels, writes her line; a Godot avatar performs it with voice and lip sync. Every run replays exactly from its log, and 26 scripted chat scenarios test her behaviour (write-up).

SKILLS

Programming	Python, C and C++, TypeScript and JavaScript, SQL, Bash, GDScript
ML, vision	PyTorch, TensorRT, ONNX, CuPy, NumPy, pandas, scikit-learn, XGBoost, OpenCV
LLM systems	tool-using assistants, structured output, scenario-based evaluation, llama.cpp
Planning	discrete-event simulation, scheduling, OR-Tools CP-SAT, reinforcement learning
Linux, graphics	X11/XCB, D-Bus, systemd, udev, OpenGL/EGL, CUDA-OpenGL interop, PulseAudio, GStreamer, CMake
Embedded	Raspberry Pi and Pico, ESP8266, Arduino, CANopen, Modbus RTU, CRC serial protocols
Web, tools	FastAPI, Flask, React, Svelte, SQLite, MySQL, Git, uv, Blender, Godot
Languages	Turkish (native); English (professional; degree taught in English)

INTERESTS

Learning theory, cognitive science, philosophy of mind and of the self, theory of computation.